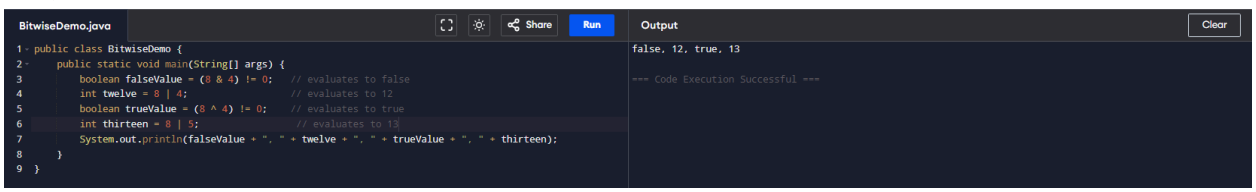
Activity #2 – Operators

General Instructions:

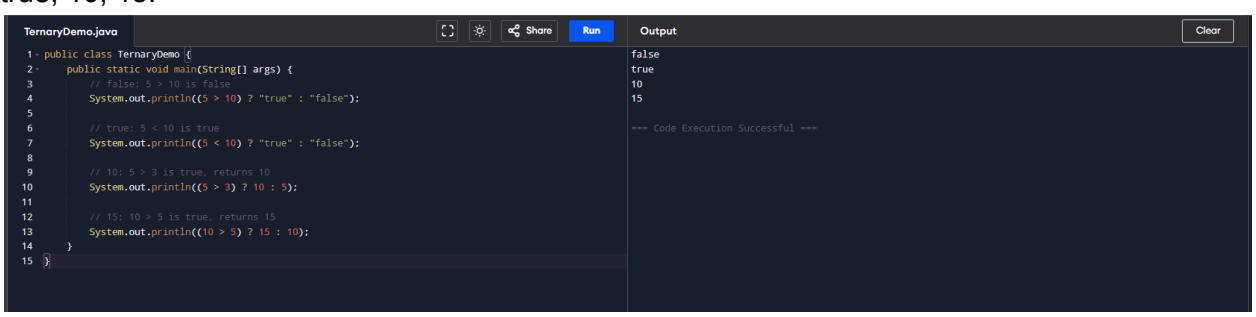
1. Create a Java program based on the questions.
2. Follow the exact required output shown in the question.
3. You must use the specific operator required in each number.
4. Do NOT hardcode the final answers (e.g., System.out.println(86); without computation is not allowed).
5. Show proper use of variables and correct operator usage.
6. Follow proper Java syntax, indentation, and naming conventions.
7. Use System.out.println() for displaying output.
8. Compile and run your program to verify correctness before submission.
9. Submission Format: Section_LastName_FirstName_OperatorsActivity#2.pdf

Please answer the following questions:

1. Create a Java program using bitwise operators that will produce the output: false, 12, true, 13.



2. Create a Java program using the ternary operator that will produce the output: false, true, 10, 15.



3. Create a Java program using assignment operators that will produce the output: Addition: 86, Subtraction: 104, Multiplication: 200, Division: 25.

AssignmentDemo.java

```java
public class AssignmentDemo {
    public static void main(String[] args) {
        // Addition: 60 += 26 → 86
        int add = 60;
        add += 26;
        System.out.println("Addition: " + add);

        // Subtraction: 200 -= 96 → 104
        int sub = 200;
        sub -= 96;
        System.out.println("Subtraction: " + sub);

        // Multiplication: 20 *= 10 → 200
        int mul = 20;
        mul *= 10;
        System.out.println("Multiplication: " + mul);

        // Division: 100 /= 4 → 25
        int div = 100;
        div /= 4;
        System.out.println("Division: " + div);
    }
}
```

Output

```
Addition: 86
Subtraction: 104
Multiplication: 200
Division: 25

=== Code Execution Successful ===
```

4. Create a Java program using arithmetic, relational, logical, and unary operators that will produce the output: TRUE, TRUE, FALSE, 20, FALSE, FALSE, TRUE, 60.

OperatorDemo.java

```java
public class OperatorDemo {
    public static void main(String[] args) {
        // TRUE: 15 > 10 (relational)
        System.out.println((15 > 10) ? "TRUE" : "FALSE");

        // TRUE: 8 * 3 == 24 (arithmetic + relational)
        System.out.println((8 * 3 == 24) ? "TRUE" : "FALSE");

        // FALSE: 10 + 5 < 10 (arithmetic + relational)
        System.out.println((10 + 5 < 10) ? "TRUE" : "FALSE");

        // 20: 40 / 2 (arithmetic)
        System.out.println(40 / 2);

        // FALSE: 25 == 30 (relational)
        System.out.println((25 == 30) ? "TRUE" : "FALSE");

        // FALSE: !(10 > 5) (unary + relational)
        System.out.println(!(10 > 5) ? "TRUE" : "FALSE");

        // TRUE: 15 < 20 && 10 < 15 (logical + relational)
        System.out.println((15 < 20 && 10 < 15) ? "TRUE" : "FALSE");

        // 60: 20 * 3 (arithmetic)
        System.out.println(20 * 3);
    }
}
```

Output

```
TRUE
TRUE
FALSE
20
FALSE
FALSE
TRUE
60

=== Code Execution Successful ===
```